Customer Shopping Behaviour Analysis

1. Project Overview

The project analyses customer behaviour using transactional data from 3,900 purchases across various product categories. The goal is to uncover insights into meaningful patterns, customer segments, product preferences, and subscription behaviour to guide strategic business decisions.

2. DataSet Summary

--Rows:3900

--Columns:18

Key Features:

-Customer demographic (Age, Gender, Location, Subscription Status)

-Purchase details (Item purchased, Category, Purchase Amount, Season, Size, Color)

-Shopping behaviour (Discount applied, Promo Code Used, Previous Purchases , Frequency of Purchases, Review rating, Shipping Type)

-Missing values: 37 values in Review Rating Column

3.EDA using Python

Step 1: Data Loading

Load and import the dataset using pandas.

```
df.head()
```

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discount Applied	Promo Code Used	Previous Purchases	Paym Meth
0	1	55	Male	Blouse	Clothing	53	Kentucky	L	Gray	Winter	3.1	Yes	Express	Yes	Yes	14	Ver
1	2	19	Male	Sweater	Clothing	64	Maine	L	Maroon	Winter	3.1	Yes	Express	Yes	Yes	2	C
2	3	50	Male	Jeans	Clothing	73	Massachusetts	S	Maroon	Spring	3.1	Yes	Free Shipping	Yes	Yes	23	Cr C
3	4	21	Male	Sandals	Footwear	90	Rhode Island	M	Maroon	Spring	3.5	Yes	Next Day Air	Yes	Yes	49	Pa
4	5	45	Male	Blouse	Clothing	49	Oregon	M	Turquoise	Spring	2.7	Yes	Free Shipping	Yes	Yes	31	Pa

Step 2 : Initial Data Understanding:

To understand the structure and quality of the dataset, the following functions were used:

- df.head() → To view first few rows
- df.info() → To check data types and non-null counts
- df.describe(include="all") → To get statistical summary
- df.isnull().sum() → To detect missing values

Step 3: Handling Missing Values (Data Cleaning)

Missing values were found in the **Review Rating** column.

Instead of removing rows, missing values were filled using **median imputation based on category**.

Step 4: **Case Snaking (Column Name Standardization)**

Column names were standardized by:

- Converting all column names to lowercase
- Replacing spaces with underscores

Step 5: Feature Engineering

New meaningful features were created to improve analysis.

- ❖ A new column **age group** was created using quartile-based segmentation. For this purpose we use “pd.qcut( )” function.
- ❖ A new numerical column **purchase_frequency_days** was created by mapping textual frequency into number of days.

Example:

- Weekly → 7
- Monthly → 30
- Quarterly → 90

Step 6: Removing Duplicate Information (Data Reduction)

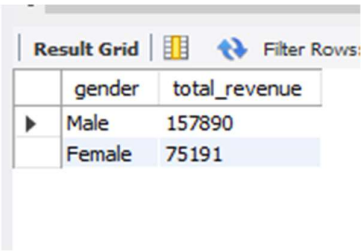
It was found that both columns contained identical values.

Therefore, one redundant column (promo_code_used) was dropped.

3.Data Analysis using SQL

We peformed structred analysis in MYSQL to answer key business questions:

Revenue by Gender: What is the total revenue generated by male vs female customer?



High-Spending Discount Users: Which customers used a discount but still spent more than the average purchase amount?

Result Grid | Filter Rows:

customer_id	purchase_amount
2	64
3	73
4	90
7	85
9	97
12	68
13	72
16	81
20	90
22	62
24	88

Top 5 Products by Rating: Which are the top 5 products with the highest average review rating?

item_purchased	Avg Product Ratimg
Gloves	3.86
Sandals	3.84
Boots	3.82
Hat	3.8
Skirt	3.78

Shipping: Compare the average Purchase Amounts between Standard and Express Shipping.

Result Grid | Filter Rows:

shipping_type	Avg Purchase Amount
Express	60.48
Standard	58.46

Subscriber vs Non-subscriber: Do subscribed customers spend more? Compare average spend and total revenue between subscribers and non-subscribers?

Result Grid | Filter Rows: | Export: | Wrap Cell C

subscription_status	Total Customers	AVG_SPEND	Total_Revenue
Yes	1053	59.49	62645
No	2847	59.87	170436

Top 5 products with discount: Which 5 products have the highest percentage of purchase with discount applied?

item_purchased	discount_rate
Hat	50.00
Sneakers	49.66
Coat	49.07
Sweater	48.17
Pants	47.37

New,Returning,Loyal: Segment customers into New, Returning, and Loyal based on their total number of previous purchase and show the count of each segment?

Result Grid | Filter Rows: | Expo

customer_segment	NO. OF CUSTOMERS
Loyal	3116
Returning	701
NEW	83

Top 3 most purchased products in Each Category:

item_rank	category	item_purchased	total_orders
1	Accessories	Jewelry	171
2	Accessories	Sunglasses	161
3	Accessories	Belt	161
1	Clothing	Blouse	171
2	Clothing	Pants	171
3	Clothing	Shirt	169
1	Footwear	Sandals	160
2	Footwear	Shoes	150
3	Footwear	Sneakers	145
1	Outerwear	Jacket	163
2	Outerwear	Coat	161

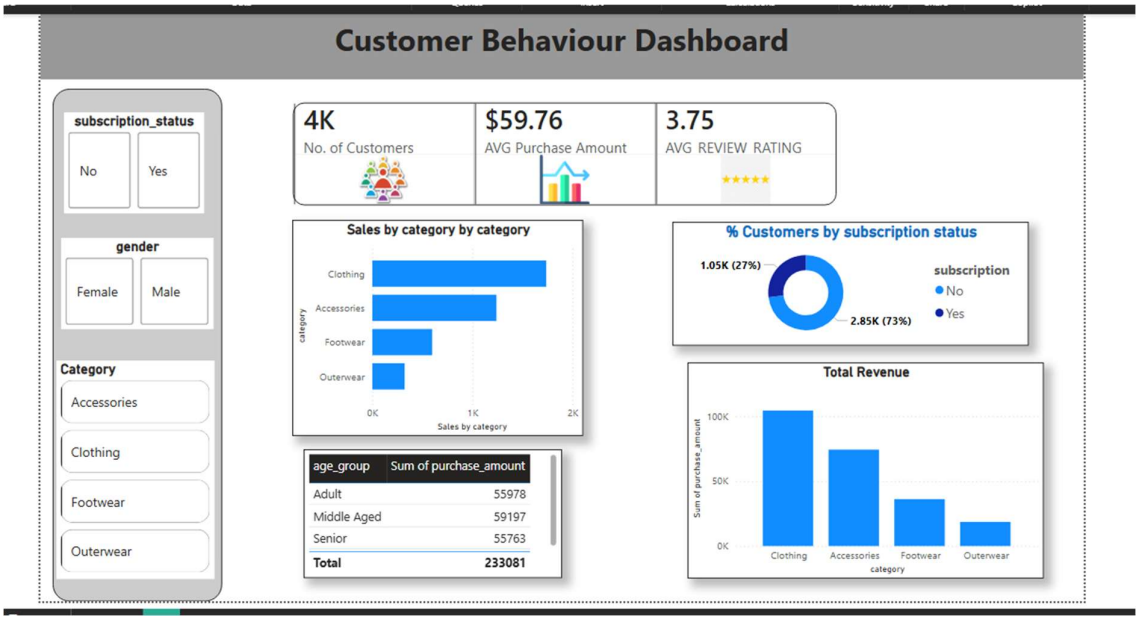
Repeat buyers: Are customers who are repeat buyers (more than 5 previous purchases) also likely to subscribe?

subscription_status	repeat_buyers
Yes	958
No	2518

Age Group: What is the revenue contribution of each group?

age_group	total_revenue
Young Adult	62143
Middle Aged	59197
Adult	55978
Senior	55763

5.Dashboard in PowerBi



6.Business Recommendation

- **Boost Subscription**: Promote exclusive benefits for subscribers.
- **Customer Loyalty Programs**: Reward repeat buyers to move them into “Loyal” segment.
- **Review Discount Policy**: Balance sales boosts with margin control.
- **Product Positing**: Highlight top-rated and best-selling products in campaigns.
- **Targeted Marketing**: Focus efforts on high-revenue age groups and express-shipping users.